



Risk-Sensitive Linear/Quadratic/Gaussian Control

Author(s): P. Whittle

Reviewed work(s):

Source: *Advances in Applied Probability*, Vol. 13, No. 4 (Dec., 1981), pp. 764-777

Published by: [Applied Probability Trust](#)

Stable URL: <http://www.jstor.org/stable/1426972>

Accessed: 22/02/2013 11:06

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at
<http://www.jstor.org/page/info/about/policies/terms.jsp>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



Applied Probability Trust is collaborating with JSTOR to digitize, preserve and extend access to *Advances in Applied Probability*.

<http://www.jstor.org>

RISK-SENSITIVE LINEAR/QUADRATIC/GAUSSIAN CONTROL

P. WHITTLE,* *University of Cambridge*

Abstract

The conventional linear/quadratic/Gaussian assumptions are modified in that minimisation of the expectation of cost G defined by (2) is replaced by minimisation of the criterion function (5). The scalar $-\theta$ is a measure of risk-aversion. It is shown that modified versions of certainty equivalence and the separation theorem still hold, that optimal control is still linear Markov, and state estimate generated by a version of the Kalman filter. There are also various new features, remarked upon in Sections 5 and 7. The paper generalises earlier work of Jacobson.

LINEAR/QUADRATIC CONTROL; RISK-SENSITIVITY; CERTAINTY EQUIVALENCE; KALMAN FILTER

1. Introduction

The term LOG is an abbreviation for linear/quadratic/Gaussian, and is fairly standard as a description of a model of control optimisation in which one assumes linear dynamics, quadratic cost function and Gaussian noise (see Kushner (1971), Davis (1977), Whittle (1982)). Our basic assumptions are the standard LOG assumptions, but for a generalisation in the criterion. The generalisation adds a significant new parameter to the formulation, measuring the optimiser's sensitivity to risk. All the usual concepts survive in this extension (optimality of linear Markov control, Riccati recursions, certainty equivalence and the Kalman filter) although with significant modifications.

So, we take the time variable t as discrete, and assume the plant equation

$$(1) \quad x_{t+1} = Ax_t + Bu_t + \alpha_t + \varepsilon_t$$

where x_t is the vector state variable, u_t the vector control variable, α_t a completely known disturbance term and ε_t is process noise, whose properties we shall specify shortly. We assume a T -horizon situation, with cost function

$$(2) \quad G = \sum_{t=0}^{T-1} (x_t' R x_t + u_t' Q u_t) + x_T' \Pi_T x_T.$$

Received 3 September 1980; revision received 8 January 1981.

* Postal address: Statistical Laboratory, University of Cambridge, 16 Mill Lane, Cambridge CB2 1SB, U.K.

Let us denote the state history $(x_0, x_1, \dots, x_t)$ to time t by X_t , with a similar convention for other variables. The control at time t must be chosen as a function of the observation history W_t at time t . In the case of complete state observation this is $W_t = (X_t, U_{t-1}, t)$. In the case of incomplete state observation this is rather $W_t = (y_t, W_0, U_{t-1}, t)$, where y_t is the vector of new observations on states which become available at time t , and W_0 is the initial information available on x_0 . It is common to suppose that x_0 is known to be normally distributed with mean $\hat{x}_0$ and covariance matrix V_0 , and that the observation/state relation is of the form

$$(3) \quad y_{t+1} = Cx_t + \eta_t.$$

Here η_t is observation noise and $\{(\varepsilon_t, \eta_t)\}$ is a Gaussian white noise process. We shall assume this, with the simplifying assumption that (ε_t, η_t) have joint covariance matrix

$$(4) \quad \text{Cov}(\varepsilon_t, \eta_t) = \begin{pmatrix} N & \cdot \\ \cdot & M \end{pmatrix}.$$

That is, we assume the process and observation noise processes independent, with constant covariance matrices. To avoid discussion of degenerate cases we shall assume the matrices R, Q, N and M positive definite.

The most general LOG model at this level assumes a cross-penalty term $2u_t' Sx_t$, under the sum in (2), a cross-covariance L between ε_t and η_t in (4), and that the matrices $A, B, C, R, Q, S, L, M, N$ may all be time-dependent. This degree of generality is as easily accommodated in the treatment we shall give, but we consider the simpler case for simplicity of exposition. The cost function (2) penalises deviations of x and u from 0; the case where one penalises deviations from prescribed time-varying values can be normalised to this by a redefinition of x, u and α .

The usual criterion is that one chooses controls (i.e. chooses the dependence of u_t upon W_t) so as to minimise $\mathbb{E}(G)$. We shall rather choose to minimise

$$(5) \quad \gamma(\theta) = -2\theta^{-1} \log \mathbb{E}(e^{-\frac{1}{2}\theta G})$$

where θ is a real scalar. The conventional criterion is $\gamma(0)$, in that $\mathbb{E}(G)$ is the limit of $\gamma(\theta)$ as θ converges to 0 from either side.

In fact, if $\theta \text{Var}(G)$ is small then $\gamma(\theta) \sim \mathbb{E}(G) - \frac{1}{4}\theta \text{Var}(G)$, which illustrates in a local kind of way that the cases $\theta = 0$, $\theta > 0$ and $\theta < 0$ correspond respectively to what an economist would term *risk-neutral*, *risk-preferring* and *risk-averse* attitudes on the part of the optimiser (see DeGroot (1970), p. 99 or Bertsekas (1976), p. 17). In the case $\theta > 0$ one is maximising the expectation of a convex decreasing function of G . For a given value of $\mathbb{E}(G)$ the introduction

of statistical variation in G improves the value of the criterion, because the occurrence of some values of G smaller than $E(G)$ is regarded as more than outweighing the necessary occurrence of some values greater than $E(G)$. In the case $\theta < 0$ one is minimising the expectation of a convex increasing function of G , and takes the opposite evaluation: the penalty inherent in the occurrence of values of G greater than $E(G)$ outweighs the alleviation in penalty caused by the necessary occurrence of some values less than $E(G)$.

We shall refer to the cases $\theta > 0$ and $\theta < 0$ as the *optimistic* and *pessimistic* cases respectively, as it will turn out that the form of the criterion leads one to act as though all uncertainties were likely to turn out to one's advantage or disadvantage according as $\theta > 0$ or $\theta < 0$. It will be a matter of interest to determine how far optimism or pessimism can be pushed before a crisis of some kind occurs (see Section 5).

The LQG problem with criterion (5) was proposed and analysed by Jacobson (1973), (1977). In his penetrating 1973 paper Jacobson treated the case of perfect state observation, and derived essentially the material of our Sections 4 and 5, although with a rather different form for the Riccati recursions. In his 1977 work he considered the case of imperfect state observation, but could find no analogue of the familiar risk-neutral results, and could develop only an infinite-order recursion for state estimates. He attributed the apparent lack of separation and symmetry (between control optimisation and state estimation) to the seeming absence of a certainty equivalence property. However, we shall show that all desirable features of the risk-neutral treatment persist in the risk-sensitive case, with appropriate definition. We show that a certainty equivalence property holds (Section 2), that the problems of control optimisation and state estimation can be decoupled (Section 3), solved separately (Sections 4 and 6) and then coupled by appeal to the certainty equivalence principle (Section 7). All the familiar results survive, including generation of state estimates by a first-order linear recursion: the Kalman filter. Indeed, the approach provides a treatment which is remarkably concise and inclusive by the standards of conventional treatments for the risk-neutral case.

We find it convenient to work with modified versions of the Riccati recursions which show transparently the effect of risk-sensitivity: to modify the control-power matrix $BQ^{-1}B'$ and the observation information matrix $C'M^{-1}C$ to $BQ^{-1}B' + \theta N$ and $C'M^{-1}C + \theta R$ respectively.

Repeated appeal is made to certain standard results concerning the maximisation and integration of Gaussian densities. For convenience these are summarised.

Lemma. Let $S(u, v, w)$ be a positive definite quadratic form in the components of vectors u and v , where w is a fixed parameter. Suppose this attains its

minimum at $u = u^*$, $v = v^*$. Then

$$\max_u \int e^{-S(u,v,w)} dv \propto e^{-S(u^*,v^*,w)}$$

and the maximum is attained at $u = u^*$. More explicitly,

$$\int e^{-S(u,v,w)} dv = (2\pi)^{\frac{1}{2}n} |S_{vv}|^{-\frac{1}{2}} e^{-S(u^*,v^*,w)} e^{-\frac{1}{2}T}$$

where $n = \dim(v)$, $T = (u - u^*)'[S_{uu} - S_{uv}S_{vv}^{-1}S_{vu}](u - u^*)$, and $\begin{pmatrix} S_{uu} & S_{uv} \\ S_{vu} & S_{vv} \end{pmatrix}$ is the partitioned matrix of the quadratic form.

2. Stress; certainty equivalence

For later convenience let us define the quadratic forms

$$r_t = x_t' R x_t$$

$$q_t = u_t' Q u_t$$

$$n_t = (x_{t+1} - A x_t - B u_t - \alpha_t)' N^{-1} (x_{t+1} - A x_t - B u_t - \alpha_t)$$

$$m_t = (y_{t+1} - C x_t)' M^{-1} (y_{t+1} - C x_t)$$

so that the cost function G can be written

$$(6) \quad G = \sum_0^{T-1} (r_t + q_t) + x_T' \Pi_T x_T$$

and the residual sum of squares (the quadratic function in the exponent of the joint density of X_T, Y_T) can be written

$$(7) \quad D = (x_0 - \hat{x}_0)' V_0^{-1} (x_0 - \hat{x}_0) + \sum_0^{T-1} (n_t + m_t).$$

Define the *total stress* as

$$(8) \quad S = G + \theta^{-1} D.$$

We can regard G as a component of stress due to deviation of x and u from desired values; D as a component of stress due to discrepancy between values of the various variables x, y and u . These components are combined positively in the case $\theta > 0$, differentially in the case $\theta < 0$.

Let us say that a variable renders S *extreme* if it minimises S in the case $\theta > 0$, maximises S in the case $\theta < 0$.

Theorem 1 (Certainty equivalence). Suppose the stress S freely minimised with respect to $u_t, u_{t+1}, \dots, u_{T-1}$ and freely extremised with respect to $x_0, x_1, \dots, x_T$

and $y_{t+1}, y_{t+2}, \dots, y_T$ for the given value of W_t . The order in which these operations are applied is irrelevant, and the value of u_t thus determined is the optimal value of control at time t .

In the case $\theta < 0$ these calculations have a meaning only if S is negative definite in the extremising variables.

This theorem generalises the familiar statement of certainty equivalence in the case $\theta = 0$ which might be expressed as follows (see Whittle (1963), p. 137). Let G be written as a function $G(u, \zeta)$ of control sequence u and the various random variables ζ . If $G(u, E(\zeta | W_t))$ is minimised freely with respect to $u_t, u_{t+1}, \dots, u_{T-1}$ then the value of u_t thus determined is the optimal value of control at time t . Note that $E(\zeta | W_t)$ is the value of ζ minimising D .

In the present case the D -minimising values of unobservables at time t are replaced by S -extremising values. These values are to be regarded as the estimates of current unobservables which extremise total stress. The values of $u_{t+1} \dots u_{T-1}$ determined in the theorem chosen to minimise total stress are likewise current estimates of optimal future decisions.

Proof. Let us take the argument initially in the case $\theta > 0$. Using f to denote probability density, let us define

$$(9) \quad G_t(W_t) = f(W_t | W_0) E(e^{-\frac{1}{2}\theta G} | W_t)$$

where the expectation is calculated on the assumption that $u_t, u_{t+1}, \dots, u_{T-1}$ have been optimally determined. The aim is then to evaluate $G_0(W_0)$, and with it the optimal control for all t . We shall formulate the optimisation recursively, using the methods of dynamic programming. For $t = T$

$$(10) \quad \begin{aligned} G_T(W_T) &\propto \int e^{-\frac{1}{2}\theta S} dX_T \\ &\propto \exp \left[-\frac{1}{2}\theta \min_{x_0, x_1, \dots, x_T} S \right] \end{aligned}$$

the second relation following from the lemma of Section 1. It follows from dynamic programming arguments (see e.g. Kreps (1977), Whittle (1982)) that

$$(11) \quad G_t(W_t) = \max_{u_t} \int G_{t+1}(W_{t+1}) dy_{t+1}$$

where the maximisation is free, and that the value of u_t thus determined is the optimal control at time t .

If we set $G_t(W_t) = \exp [-\frac{1}{2}\theta \phi_t(W_t)]$ then it follows from (10) and the lemma that $\phi_T(W_T)$ is a positive definite quadratic form in Y_T, U_{T-1} . Appeal to (11) for $t = T-1, T-2, \dots$ and the lemma then establishes recursively that $\phi_t(W_t)$ is a positive definite quadratic form in Y_t, U_{t-1} and that the recursion (11)

amounts to

$$(12) \quad \phi_t(W_t) = \text{const.} + \min_{u_t, y_{t+1}} \phi_{t+1}(W_{t+1}),$$

the ‘constant’ being a function of t alone. From this and the effective specification of ϕ_T in (10) we deduce the theorem in the case $\theta > 0$.

In the case $\theta < 0$ the same argument holds except that u_t is minimised out in (11), y_{t+1} is maximised out in (12), and $x_0, x_1, \dots, x_T$ maximised out in (10). However, the integrals with respect to unobservables in (10), (11) will be convergent, and evaluated by maximisation as in (10), (12), if and only if S is negative definite in these unobservables.

If $\theta > 0$ then the order of the minimisations can certainly be interchanged. If $\theta < 0$ and the negative-definiteness condition is fulfilled, then we are dealing with quadratic forms possessing a saddlepoint. The operations of min and max commute under these conditions.

3. Past and future stresses

Let us define the past stress (at time t) as

$$(13) \quad P(x_t, W_t) = \text{ext}_{x_0, \dots, x_{t-1}} \left[\theta^{-1}(x_0 - \hat{x}_0)' V_0^{-1}(x_0 - \hat{x}_0) + \sum_{s=0}^{t-1} (r_s + q_s + \theta^{-1}(n_s + m_s)) \right]$$

and the future stress as

$$(14) \quad F_t(x_t) = \min_{u_t, \dots, u_{T-1}} \text{ext}_{x_{t+1}, \dots, x_T} \text{ext}_{y_{t+1}, \dots, y_T} \left[\sum_{s=t}^{T-1} (r_s + q_s + \theta^{-1}(n_s + m_s)) + x_T' \Pi_T x_T \right] \\ = \min_{u_t, \dots, u_{T-1}} \text{ext}_{x_{t+1}, \dots, x_T} \left[\sum_{s=t}^{T-1} (r_s + q_s + \theta^{-1} n_s) + x_T' \Pi_T x_T \right].$$

Here ‘ext’ stands for ‘extremum’ and so must be read as ‘min’ or ‘max’ according as $\theta > 0$ or $\theta < 0$. These quantities can be regarded as the components of total stress attributable to past and future if the current state value x_t is specified, future actions are optimised out, and variables currently unobservable are extremised out. Note that the u -minimisations in (14) are free, and will automatically yield (x_t, t) -measurable evaluations.

Expressions (13) and (14) obviously obey the recursions

$$(15) \quad P_{t+1}(x_{t+1}, W_{t+1}) = \text{ext}_{x_t} [P_t(x_t, W_t) + r_t + q_t + \theta^{-1}(m_t + n_t)]$$

$$(16) \quad F_t(x_t) = \min_{u_t} \text{ext}_{x_{t+1}} [r_t + q_t + \theta^{-1}n_t + F_{t+1}(x_{t+1})]$$

with end conditions

$$(17) \quad P_0(x_0, W_0) = \theta^{-1}(x_0 - \hat{x}_0)' V_0^{-1}(x_0 - \hat{x}_0)$$

$$(18) \quad F_T(x_T) = x_T' \Pi_T x_T.$$

The forward recursion (15) revises current state information as a new observation is incorporated; the backward recursion (16) revises current action criteria as the horizon recedes by a single step.

In terms of these quantities Theorem 1 can be rewritten as follows.

Theorem 2. Let $u(x_t, t)$ be the minimising value of u_t in (16), and let $\bar{x}_t$ be the value of x_t extremising $P_t(x_t, W_t) + F_t(x_t)$, a function of W_t . Then the optimal value of u_t is $u(\bar{x}_t, t)$.

This is indeed no more than a rephrasing of Theorem 1, as the reader can verify. However, it does bring out certainty-equivalence in another way. If state is completely observed, i.e. if $W_t = (X_t, U_{t-1}, t)$, then the optimal control is just $u_t = u(x_t, t)$. The optimal control in the case of imperfect state observation is obtained just by replacing x_t by its minimum-stress estimate $\bar{x}_t$, which is a version of certainty equivalence.

4. Complete state observation

In the case of complete state observation the optimal control rule is fully determined by solving the recursion (16) for future stress. For simplicity we shall consider the homogeneous case $\alpha_t = 0$ for the moment, and return to the deterministically-disturbed case in Section 8.

We shall define the matrix $J = BQ^{-1}B'$ and shall use the convention of writing $U > V$ or $U \geq V$ if $U - V$ is positive definite or non-negative definite respectively, where U and V are symmetric matrices of equal order.

Theorem 3. The future stress is quadratic in the state variable

$$(19) \quad F_t(x_t) = x_t' \Pi_t x_t,$$

and the optimal control is linear:

$$(20) \quad u(x_t, t) = K_t x_t.$$

The matrix Π_t obeys the Riccati recursion

$$(21) \quad \Pi_t = R + A'(J + \theta N + \Pi_{t+1}^{-1})^{-1}A$$

and K_t has the evaluation

$$(22) \quad K_t = -Q^{-1}B'(J + \theta N + \Pi_{t+1}^{-1})^{-1}A.$$

For these assertions to be true in the case $\theta < 0$ it is necessary that $\Pi_s < -(\theta N)^{-1}$ ($s > t$).

Proof. The argument is the familiar inductive one, but there are a few points that need to be brought out. Relation (19) is certainly true at T . Assuming it true at $t+1$ we have then

$$(23) \quad F_t(x) = \min_u \text{ext}_\varepsilon [x'Rx + u'Qu + \varepsilon'N^{-1}\varepsilon/\theta + (Ax + Bu + \varepsilon)' \Pi_{t+1} (Ax + Bu + \varepsilon)].$$

Consider first the case $\theta > 0$. Then it is apparent from (23) that it is as though ε were acting as an extra component of control, and one can recover the treatment of this case from the conventional one for $\theta = 0$ by the substitutions

$$u \rightarrow \begin{pmatrix} u \\ \varepsilon \end{pmatrix} \quad B \rightarrow [B \quad I] \quad Q \rightarrow \begin{pmatrix} Q \\ \cdot \\ (\theta N)^{-1} \end{pmatrix}.$$

With these substitutions in the conventional treatment one can obtain relations for Π and K . The relations, are, however, not quite in the form (21), (22) which is natural and convenient in the present case. To recover these we write (23) rather as

$$(24) \quad F_t(x_t) = \min_u \text{ext}_\varepsilon \max_z [x'Rx + u'Qu + \varepsilon'N^{-1}\varepsilon/\theta + 2z'(Ax + Bu + \varepsilon) - z'\Pi_{t+1}^{-1}z]$$

where z can in fact be identified as the conjugate variable $\frac{1}{2}(\partial F_{t+1}/\partial x_{t+1})$. Writing down conditions for stationarity with respect to u , ε and z , solving these for u , ε in terms of z , and then solving for z , one obtains evaluations (21), (22). These are also valid in the case $\theta < 0$, provided only that the bracketed expression in (23) is indeed then negative definite in ε , i.e. that $\Pi_{t+1} + (\theta N)^{-1} < 0$.

As noted from (23) in the case $\theta > 0$, it is as though ε were available to take over some of the role of control, so that actual control u could be chosen more moderately than in the case $\theta = 0$. This is the optimistic view; that process noise will be such as to direct the state somewhat in the desired direction. This fact is illustrated by formulae (21), (22), which differ from the corresponding

formulae for the risk-neutral case by the substitution

$$(25) \quad J \rightarrow J + \theta N.$$

The matrix J can be regarded as a measure of 'control power'; this is now effectively increased by a multiple of the process noise power matrix.

In the case $\theta < 0$ one has the contrary situation: the extremising ε in (23) will be such as to increase the last term, and one is effectively making the pessimistic assumption that process noise will be such as to direct state in an undesired direction. Correspondingly, formula (25) now reflects a decrease in effective control power.

5. Instability and breakdown

As $\theta \rightarrow \pm\infty$ the optimiser's attitude moves to an extreme of optimism or pessimism, and it is a question of how far it can move before control breaks down. In the one case, the attitude that uncertainties work to one's advantage may lead to a control which is too lax for stability. In the other, the attitude that uncertainties work to one's disadvantage may lead to a conviction of inability to control, and paralysis. These ideas are best considered in the stationary infinite horizon limit, obtained either by holding t fixed as $T \rightarrow +\infty$ or by holding $T = 0$ as $t \rightarrow -\infty$.

It follows by established arguments (see e.g. Kushner (1971), Bertsekas (1976), Whittle (1982)) that, under appropriate conditions infinite-horizon limits exist. This can be expressed by saying that the sequence Π_t generated by (21) has a unique finite positive definite limit Π as $t \rightarrow -\infty$, for any $\Pi_0 \geq 0$. Sufficient conditions are, for example, that $R > 0$, $J + \theta N \geq 0$, and the system is controllable.

Furthermore, consider the matrix Γ_t defined by

$$(26) \quad Ax_t + B\hat{u} + \hat{\varepsilon} = \Gamma_t x_t$$

where $\hat{u}, \hat{\varepsilon}$ are the extremal values obtaining in (23). Then Γ_t also has a limit, verified as $\Gamma = (I + (J + \theta N)\Pi)^{-1}A$, which is a stability matrix. That is, all its eigenvalues lie strictly inside the unit circle.

However, in the limiting case $\Pi_t = \Pi$, $\Gamma_t = \Gamma$, $K_t = K$ the actual dynamic equation that the controlled system follows is

$$(27) \quad x_{t+1} = \Gamma^* x_t + \varepsilon_t$$

where $\Gamma^* x_t = Ax_t + B\hat{u}_t = (A + BK)x_t$. Hence

$$(28) \quad \Gamma^* = (I + \theta N\Pi)\Gamma.$$

For stability of the controlled process one would require that Γ^* be stable. In

the case $\theta > 0$ one might suspect from (28) that it need not be if θ were large enough, and this turns out in fact to be the case. In the case $\theta < 0$ one has $\Lambda := I + \theta N^{\frac{1}{2}} \Pi N^{\frac{1}{2}} < I$ and so certainly has stability as long as $\Lambda \geq 0$, but the situation is unclear if $\Lambda \not\geq 0$. In fact, the solution is unacceptable on other grounds in this latter case.

As θ becomes large and positive, corresponding to extreme optimism, then one type of crisis is possible. As θ becomes large and negative, corresponding to extreme pessimism, then another type of crisis is certain. In the first case, if the uncontrolled system is unstable, then the controlled system will also become so for θ large; the optimiser is controlling too weakly, in the optimistic belief that process noise will do his work for him. In the second case, when θ is sufficiently large (in the negative direction) the optimiser's pessimistic assessment of the value of ε so outweighs his knowledge of its statistical distribution that expected costs become infinite: a type of neurotic breakdown.

Consider the case $\theta > 0$. Equation (21) suggests that $\Pi \rightarrow R$ as $\theta \rightarrow +\infty$, and hence that $K \rightarrow 0$, $\Gamma^* \rightarrow A$. That is, that the controlled system approaches the uncontrolled system arbitrarily closely as $\theta \rightarrow +\infty$, with the certainty of ultimate instability if the uncontrolled system is unstable.

Consider the case $\theta < 0$. We know from Theorem 3 that it is not sufficient that $\{\Pi_t\}$ should converge, but it is also necessary that $\Pi_t < -(\theta N)^{-1}$ at all stages. If the inequality is violated for some t then expected future costs at s are infinite for $s < t$. Let us write recursion (21) as $\Pi_t = \psi(\Pi_{t+1})$. Let $\mathcal{P}$ be the set of non-negative definite Π such that $\Pi \in \mathcal{P}$ implies that $\Pi \leq -(\theta N)^{-1}$ and $\psi(\Pi) \in \mathcal{P}$ (for a prescribed value of θ). Then convergence of Π_t to Π will be meaningful only if $\mathcal{P}$ is non-empty and $\Pi_0 \in \mathcal{P}$, when also $\Pi \in \mathcal{P}$. In the case $\theta < 0$ these conditions must be added to those stated at the beginning of the section.

$\mathcal{P}$ is certainly non-empty for $\theta = 0$. We see from (21) that, for prescribed Π_{t+1} , then $\Pi_t \rightarrow R$ as $\theta \rightarrow -\infty$, and so $\Pi_t < -(\theta N)^{-1}$ for θ sufficiently large and negative. That is, for θ sufficiently large and negative the condition $\Pi_t < -(\theta N)^{-1}$ will be violated for some t , and breakdown occurs.

Identity (30) below is interesting in this connection. We see from (23) that

$$(29) \quad x' \Pi x = \underset{\varepsilon}{\text{ext}} [x' U x + \theta^{-1} \varepsilon' N^{-1} \varepsilon + (\Gamma^* x + \varepsilon)' \Pi (\Gamma^* x + \varepsilon)]$$

where $U = R + K' Q K$. From (29) it follows that

$$\mathbb{E} [\exp -\frac{1}{2} \theta [x'_0 U x_0 + x'_1 \Pi x_1]] = \lambda \exp [-\frac{1}{2} \theta x'_0 \Pi x_0]$$

and hence that

$$(30) \quad \mathbb{E} \left[\exp -\frac{1}{2} \theta \left[\sum_{s=0}^{t-1} x'_s U x_s + x'_t \Pi x_t \right] \right] = \lambda^t \exp [-\frac{1}{2} \theta x'_0 \Pi x_0]$$

where

$$(31) \quad \lambda = |I + \theta N \Pi|^{-\frac{1}{2}}.$$

The expectation in (30) is conditional on the value of x_0 , and calculated on the assumption that the process is following its limiting controlled form (27).

Relation (30) seems to argue a kind of equilibrium, and so of stability. However, such is not the case: Equation (30) is compatible with an unstable Γ^* . On the other hand, evaluation (31) does show a sensitivity to breakdown; it becomes infinite if θ is negative and large enough that $|I + \theta N \Pi| = 0$.

6. Incompletely observed state

In the case of incompletely observed state we must also solve recursion (15).

Theorem 4. $P_t(x_t, W_t)$ has the form

$$(32) \quad P_t(x_t, W_t) = \theta^{-1}(x_t - \hat{x}_t)' V_t^{-1}(x_t - \hat{x}_t) + \dots$$

where $t \dots$ indicates terms independent of x_t . The matrix V_t and vector $\hat{x}_t$ satisfy the recursions

$$(33) \quad V_{t+1} = N + A(V_t^{-1} + C'M^{-1}C + \theta R)^{-1}A'$$

$$(34) \quad \begin{aligned} \hat{x}_{t+1} &= A\hat{x}_t + Bu_t + \alpha_t \\ &+ A(V_t^{-1} + C'M^{-1}C + \theta R)^{-1}(C'M^{-1}(y_{t+1} - C\hat{x}_t) - \theta R\hat{x}_t). \end{aligned}$$

For these assertions to be true in the case $\theta < 0$ it is necessary that $V_s < -(\theta R)^{-1}$ ($s \leq t$).

In the risk-neutral case $\theta = 0$ $\hat{x}_t$ would be the least square, maximum likelihood, or conditional estimate of state $E(x_t | W_t)$ and V_t its covariance matrix. In the present case $\hat{x}_t$ is the value of state which extremises past stress; relative to $E(x_t | W_t)$ it will be biased optimistically or pessimistically according as $\theta > 0$ or $\theta < 0$. The matrix V_t^{-1} is a measure of the sharpness of this minimum.

In (33) we recognise one form of the familiar Riccati equation, and in (34) we recognise a version of the Kalman filter, updating the state estimate $\hat{x}_t$. This differs from the risk-neutral filter, partly in that the innovation $y_{t+1} - C\hat{x}_t$ and the matrix multiplying it have θ -dependent values, but also in the occurrence of the biasing term $-\theta R\hat{x}_t$. This is the risk-induced bias mentioned.

The similarity of structure in Theorems 3 and 4 is apparent, with, for example, the 'single-observation information matrix' $C'M^{-1}C$ replacing the 'control-power matrix' $J = BQ^{-1}B'$.

The effect of risk is to modify this matrix to $C'M^{-1}C + \theta R$, just as J was modified to $J + \theta N$. That is, it is as though the pressures exerted by cost-stress

supplied information (positive or negative) additional to that supplied by observations.

Proof. Relation (32) holds at $t=0$ by hypothesis; assume it to hold at t . Inserting expression (32) into (15) we see that the quadratic form holds also for $P_{t+1}(x_{t+1}, W_{t+1})$. The centring value $\hat{x}_{t+1}$ will be the value of x_{t+1} extremising $P_{t+1}(x_{t+1}, W_{t+1})$: writing down stationarity conditions for the form $P_t + r_t + q_t + \theta^{-1}(m_t + n_t)$ with respect to x_{t+1} and x_t we deduce immediately the risk-modified Kalman filter relation (34).

This form has the matrix

$$\begin{pmatrix} \beta & \gamma \\ \gamma' & \delta \end{pmatrix} = \begin{pmatrix} N^{-1} & -N^{-1}A \\ -A'N^{-1} & V_t^{-1} + C'M^{-1}C + \theta R + A'NA \end{pmatrix}$$

when regarded as a quadratic form in x_{t+1}, x_t . We shall then have, by the lemma of Section 1, $V_{t+1}^{-1} = \beta - \gamma\delta^{-1}\gamma'$. Appealing to the identity

$$(\beta - \gamma\delta^{-1}\gamma')^{-1} = \beta^{-1} + \beta^{-1}\gamma(\delta - \gamma'\alpha^{-1}\gamma)^{-1}(\gamma')^{-1}\beta^{-1},$$

obtained by comparing the alternative pivotal inversions of the partitioned matrix $\begin{pmatrix} \beta & \gamma \\ \gamma' & \delta \end{pmatrix}$, we deduce recursion (33) for V_t .

7. The coupling of state estimation and optimal control

A feature of the risk-neutral case is the separation theorem: that in the case of complete state observation the optimal control law (20) is independent of process noise characteristics, and that in the case of incomplete state observation the state estimate $\hat{x}_t$ and its covariance are unaffected by the control policy. The two are coupled by the certainty equivalence assertion: that the optimal control in the case of incomplete observation is $u_t = K_t\hat{x}_t$.

In the risk-sensitive case we do not have this complete separation: the occurrence of the term θN in the formulae of Theorem 3 means that process noise affects the optimal control rule; the occurrence of the term θR in the formulae of Theorem 4 means that state-penalties affect state estimation. Nevertheless, there is still a separation, in that the problems of determining control rule and state estimate can be decoupled by separate solution of the backward recursion (16) and the forward recursion (15). The problems can finally be coupled by application of the risk-sensitive certainty equivalence principle: Theorem 2.

We can now appeal to Theorems 2, 3 and 4 to obtain the following specific evaluation.

Theorem 5. The optimal control rule is $u_t = K_t \bar{x}_t$ where

$$(35) \quad \bar{x}_t = (I + \theta V_t \Pi_t)^{-1} \hat{x}_t$$

and $\Pi_t, K_t, V_t, \hat{x}_t$ have the evaluations asserted in Theorems 3 and 4.

The evaluation (35) of $\bar{x}_t$ will be modified in the case $\alpha_t \neq 0$; see the next section.

8. Tracking and buffering

The results of Section 4 were obtained under the supposition $\alpha_t = 0$. We shall now restore this term, so that the system is effectively buffering an injected deterministic disturbance α_t ; the tracking of a desired $\{x_t, u_t\}$ profile can be reduced to this case.

The assertions of Theorem 3 will now hold with $F_t(x_t) = (x_t - x_t^*)' \Pi_t (x_t - x_t^*) + \dots$ where Π_t is as before and $+\dots$ indicates terms independent of x_t . The term x_t^* represents the value of x_t extremising future stress, and has to be calculated. It is a quantity dual to the corresponding quantity $\hat{x}_t$ in past stress, the state estimate. One might think that one could obtain a recursion for it dual to that obeyed by $\hat{x}_t$, the Kalman filter (34), and just as easily. This is not quite true, because relation (16) involves two extremising operations, whereas relation (15) involves but one. What one finds is the following.

Theorem 6. Define $\xi_t = \Pi_t x_t^$. Then*

$$(36) \quad \xi_t = \Gamma_t' (\xi_{t+1} - \Pi_{t+1} \alpha_t)$$

where Γ_t is the matrix determined in (26):

$$\Gamma_t = (I + (J + \theta N) \Pi_{t+1})^{-1} A.$$

Relation (36) is the backward dual of the Kalman filter, although not immediately recognisable as such.

Proof. The analogue of (24) will now be

$$F_t(x) = \min_u \max_\varepsilon \max_z [x' R x + u' Q u + \varepsilon' N^{-1} \varepsilon / \theta + 2z'(A x + B u + \alpha + \varepsilon - x_{t+1}^*) - z' \Pi_{t+1}^{-1} z].$$

The value of x extremising this is x_t^* . If one writes down stationarity conditions of the bracket with respect to x, u, ε and z ; uses those for u and ε to eliminate these variables, and then solves the consequent pair of equations for x , one deduces that

$$R x_t^* + A'(J + \theta N + \Pi_{t+1}^{-1})^{-1} (A x_t^* + \alpha_t - x_{t+1}^*) = 0.$$

This reduces immediately to (36).

The solution (35) for $\bar{x}_t$ now becomes

$$\begin{aligned}\bar{x}_t &= (I + \theta V_t \Pi_t)^{-1}(\hat{x}_t + \theta V_t \Pi_t x_t^*) \\ &= (I + \theta V_t \Pi_t)^{-1}(\hat{x}_t + \theta V_t \xi_t^*).\end{aligned}$$

Note added in proof, June 1981. I have become aware of the additional references Speyer, Deyst and Jacobson (1974), Speyer (1976) and Kumar and van Schuppen (1981). These articles demonstrate that the optimal control can be generated by a finite-dimensional recursion in the cases $N=0$ and $R=0$. These restrictions are significant, in view of our finding that control-power and information matrices are effectively modified by addition of θN and θR respectively. Jacobson (1973) had of course proved finite-dimensionality of recursion in the case of perfect state observation.

That we are able to transfer the treatment of the risk-neutral case (implying also finiteness of the dimension of the generating recursions) to the general risk-sensitive case is due to the technique of decoupling evaluation of past and future stresses for prescribed current state x_t , and then later re-coupling these evaluations by appeal to the certainty equivalence assertion of Theorem 2.

References

- BERTSEKAS, D. P. (1976) *Dynamic Programming and Stochastic Control*. Academic Press, New York.
- DAVIS, M. H. A. (1977) *Linear Estimation and Stochastic Control*. Chapman and Hall, London.
- DEGROOT, M. (1970) *Optimal Statistical Decisions*. McGraw-Hill, New York.
- HOWARD, R. AND MATHESON, J. (1972) Risk-sensitive Markov decision processes. *Management Sci.* **18**, 356–369.
- JACOBSON, D. H. (1973) Optimal stochastic linear systems with exponential performance criteria and their relation to deterministic differential games. *IEEE Trans. Automatic Control* **AC-18**, 124–131.
- JACOBSON, D. H. (1977) *Extensions of Linear-Quadratic Control, Optimization and Matrix Theory*. Academic Press, New York.
- KREPS, D. M. (1972) Decision problems with expected utility criteria. I: Upper and lower convergent utility. *Math. Operat. Res.* **2**, 45–53.
- KUSHNER, H. (1971) *Introduction to Stochastic Control*. Holt, Rinehart and Winston, New York.
- WHITTLE, P. (1963) *Prediction and Regulation*. English Universities Press, London.
- WHITTLE, P. (1971) *Optimisation Under Constraints*. Wiley Interscience, London.
- WHITTLE, P. (1982) *Optimisation Over Time*. Wiley Interscience, London.

Additional references

- KUMAR, P. R. AND VAN SCHUPPEN, J. H. (1981) On the optimal control of stochastic systems with an exponential-of-integral performance index. *J. Math. Anal. Appl.* **80**, 312–332.
- SPEYER, J. L. (1976) An adaptive terminal guidance scheme based on an exponential cost criterion with application to homing missile guidance. *IEEE Trans. Automatic Control* **AC-21**, 371–375.
- SPEYER, J. L., DEYST, J. AND JACOBSON, D. H. (1974) Optimisation of stochastic linear systems with additive measurement and process noise using exponential performance criteria. *IEEE Trans. Automatic Control* **AC-19**, 358–366.